positive, neutral, negative, and very negative. In this case also, there are two algorithms:

i. The Naive Bayes algorithm trained on Janyce Wiebe's subjectivity lexicon
ii. The voter algorithm.

Readers can select any one of them.

To discuss the technicality of opinion mining with R here, I have carried out an exercise using the micro-blogger site Twitter. As mentioned earlier, the reason is obvious; Twitter comments are made up of just a little text and contain informal communication. Analysis of their sentimental value is easier than of a formal text. For instance, if we consider a formal film review, we will find a well-designed analysis of the product, but the personal, casual emotions of the writer will be absent.

Preparation

The preparation for this sentiment analysis involves two stages. The first is to install and load all packages of Twitter for Twitter JSON connectivity. After this, install and load packages related to opinion mining.

Preparation for Twitter

Twitter connectivity and reading requires the following three packages. If not already installed, install them. Then, load the libraries, if the packages have been installed properly.

```
install.packages("twitteR")
install.packages("ROAuth")
install.packages("modeest")

library(twitteR)
library("ROAuth")
library("httr")
```

Preparation for text mining and sentiment analysis

For sentiment analysis, I have downloaded the sentiment package and loaded it as shown below. The remaining relevant packages are installed and loaded as discussed earlier.

```
library(devtools)          #devtools contain load_all
load_all("sentiment")       # This package require tm and
NLP

install.packages("tm")
install.packages("plyr")
install.packages("wordcloud")
install.packages("RColorBrewer")
install.packages("stringr")
install.packages("openNLP")

library(tm)
library(plyr)
library(ggplot2)
library(wordcloud)
library(RColorBrewer)
library(sentiment)
```

Twitter connectivity

I have considered a predefined twitter application account to access the twitter space. Interested readers should create their own account to view this opinion mining experiment. You may refer earlier article from *OSFY* Jan 2018 issue.

```
download.file(url="http://curl.haxx.se/ca/cacert.
pem",destfile="cacert.pem")
cred<- OAuthFactory$new(consumerKey='HTgXiD3kqncGM93bxlBczTf
hR',
consumerSecret='djgP2zhAWKbGAgiEd4R6DXujipXRq1aTSdoD9yaHSA8q
97G80e',

requestURL='https://api.twitter.com/oauth/request_token',
accessURL='https://api.twitter.com/oauth/access_token',
authURL='https://api.twitter.com/oauth/authorize')

# enter PIN FROM https://api.twitter.com/oauth/authorize

cred$handshake(cainfo="cacert.pem")
```

This handshaking protocol requires PIN number verification, so enter your PIN as shown in the Twitter logged-in screen. Saving Twitter authentication data is a helpful step for future access to your Twitter account.

```
save(cred, file="twitter authentication.Rdata")

consumerKey='HTgXiD3kqncGM93bxlBczTfhR'
consumerSecret='djgP2zhAWKbGAgiEd4R6DXujipXRq1aTSdoD9yaHSA8q
97G80e'

AccessToken<- '1371497582-xD5GxHn
kpg8z6k0XqpnJZ3XvIyc1vVJGUsDXNWZ'
AccessTokenSecret<- Qm9tV2XvlOcwbrL2z4Qkt
A3azydtgIYPqflZglJ3D4WQ3'

setup_twitter_oauth(consumerKey, consumerSecret,AccessToken,A
ccessTokenSecret)
```

Analysis of text: Create a corpus

To perform a proper sentiment analysis, I have selected the popular 'Women's reservation' topic, and analysed the Twitter corpus to study the sentiments of all the participants with a Twitter account. To have a locality search within India, I have set the geocode at Latitude: 21.146633 and Longitude: 79.088860, within a radius of 1000 miles (Figure 1).

In total, 1500 tweets with the given keyword have been selected using the function *searchTwitter()*.